

Standard Operating Procedure

Running a Quality Improvement Project

«Hospital Name» — Supports PSQ.4.a-c

SOP No.	[Hospital to define]	Version	1.0
Effective Date	_____	Review Frequency	Annual, or on change
Prepared By Quality Coordinator Name: _____ Signature: _____			
Approved By Medical Superintendent Name: _____ Signature: _____			

1. Purpose

This SOP describes how «Hospital Name» plans, runs and documents a quality improvement (QI) project, so at least two are completed every year with a real, measurable outcome.

2. Scope

Applies to every QI project undertaken by the hospital.

3. Responsibility

Project Lead: runs the project and reports outcomes to the QI Committee.

QI Committee: selects and approves projects, tracks completion against the annual minimum of two.

4. Procedure

1. Select a project based on a high-risk, high-volume or problem-prone process, or based on indicator trends, incident analysis, or patient-reported data (see the KPI SOP for indicator trends).

2. Give the project a definite purpose, and a defined start and end date — it is a time-bound activity, not an open-ended initiative.

3. Identify which of the six domains of healthcare quality the project targets: patient safety, cost-effectiveness, patient-centredness, timeliness, efficiency, or equity in healthcare. A project should show its intended impact on at least one.

4. Measure the parameter under improvement at the start of the project (baseline) — this is what "before" is compared against at the end.

5. Run the project using a structured methodology:

— PDSA (Plan-Do-Study-Act): plan a small-scale test of a change, do it, study whether it worked, then act — adopt, adapt, or abandon the change, and repeat with the next cycle. Good for iterative, low-risk changes.

— Six Sigma: uses the DMAIC cycle (Define-Measure-Analyse-Improve-Control) to drive defects toward near-zero. Better suited to high-volume, data-rich processes.

— Lean: focuses on removing waste and non-value-adding steps. Good for process-flow and turnaround-time problems.

6. Worked example (PDSA on IV cannulation failure rate): Plan — introduce a mandatory checklist before attempting cannulation on high-risk patients; Do — trial it on one ward for two weeks; Study — compare the failure rate on that ward against baseline; Act — the rate dropped, so roll the checklist out hospital-wide as a second PDSA cycle.

7. Use analytical tools appropriate to the project — a fishbone (cause-and-effect) diagram to find causes, Pareto analysis to rank the causes by frequency and focus on the most significant few, a run chart or control chart to track the measure over time, and a flow chart to map the process being changed.

8. Measure the parameter again at the end of the project and document whether the stated goal was achieved — do not close a project without this endpoint measurement.

9. Communicate the outcome to relevant staff and, where applicable, the governing body.

5. Records

Supported by the QI Project Charter (Record 1), Domain & Impact Record (Record 2), and Methodology & Tool Log (Record 3) of the PSQ.4 Required Records pack.

6. Related Documents

Policy on Quality Improvement Projects (HCO/PSQ/POL/04) — PSQ.4 policy

7. Review

Reviewed annually, or immediately if a project reveals a gap in this procedure.

8. References

Institute for Healthcare Improvement. (n.d.). Science of Improvement: How to Improve (PDSA). <https://www.ihi.org/resources/how-to-improve>

American Society for Quality. (n.d.). Seven Basic Quality Tools. <https://asq.org/quality-resources/seven-basic-quality-tools>